



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F)

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TITLE OF THE PROJECT	DFA and NFA Simulator	
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Abstract

The DFA and NFA Simulator is an educational tool developed to demonstrate the working principles of Deterministic Finite Automata (DFA) and Non-Deterministic Finite Automata (NFA). It provides an easy way for users to define states, input symbols, transitions, start states, and final states. The system helps users understand how finite automata process input strings and determine whether they are accepted or rejected.

The simulator allows users to enter an input string and displays the sequence of state transitions performed by the DFA or NFA. For DFA, the system follows a single transition for each input symbol, while for NFA it considers multiple possible transitions. The final state is checked to determine whether the given string belongs to the language accepted by the automaton.

The project can be implemented using Python with a simple graphical user interface to make the system interactive and easy to use. It provides a visual and practical approach to understanding concepts such as finite automata, state transitions, and string validation. The simulator can be useful for students to learn and experiment with DFA and NFA concepts in Compiler Design and Formal Language Theory.

Signature of the Guide

HOD